INCH-POUND
A-A-59259E (AF)
11, March 2025
SUPERSEDING
A-A-59259D
23, May 2014

COMMERCIAL ITEM DESCRIPTION

TRUCK, AIRCRAFT REFUELING, HYDRANT SERVICING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description covers the general requirements for the R-12 Hydrant Servicing Vehicle (herein referred to as HSV) used for fuel servicing both military and commercial aircraft at Government installations worldwide. The HSV is to be capable of servicing aircraft from a Type III Hydrant System through ground servicing discharge hoses at a flow rate up to and including 750 gallons per minute (GPM), through the lift platform hoses (if equipped with a lift platform) at a flow rate up to and including 1,000 GPM, and defuel at a minimum flow rate of 80 GPM through one rear ground discharge hose without the assistance of aircraft boost pumps and 300 GPM with the assistance of aircraft boost pumps. The truck is to be air transportable in type C-130, C-17, and C-5 aircraft.

2. CLASSIFICATION.

TYPE I – Lift Platform Capability (NSN: 2320 01 125 2481)

TYPE II – No Lift Platform Capability (NSN: 2320 01 627 7046)

3. SALIENT CHARACTERISTICS.

3.1 HSV description. The HSV shall be capable of transferring fuel to and from a Type III Hydrant System when servicing military and commercial aircraft at Government installations worldwide. The HSV primary functions shall include transferring aviation jet fuel utilizing both ground servicing hoses at a flow rate up to and including 750 GPM or through both the lift platform hoses at a flow rate up to and including 1000 GPM to the aircraft from a Type III Hydrant System capable of supplying a minimum of 120 PSIG.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/

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3.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. The HSV shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The HSV shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. The HSV shall meet all applicable requirements of NFPA 407 for components that are required to be intrinsically safe when used during fuel servicing operations.

3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Metals. All metal components and piping downstream of the filtration system shall be of aluminum or stainless steel. Zinc coated metals, brass, bronze, or other copper bearing alloys shall not come in direct contact with fuel except brass components are acceptable in hose couplings/hose reducers and drain valves and copper components are acceptable in drain lines that render the product as sump fuel.

3.2.1.2 Elastomers. Elastomeric materials shall be certified compatible with jet fuel.

3.2.1.3 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. All components which come into contact with jet fuel shall be certified by their manufacturer as suitable for fuel service, corrosion resistant, and shall not be adversely affected by or affect the fuel.

3.2.1.4 Finish. The exterior finish color of the HSV shall be applied using the manufacturer's standard commercial paint system. Unless otherwise specified (see 6.1), the exterior finish color of the HSV and the inner surfaces of compartments shall be semi-gloss green, Color Number 24052 of SAE AMS-STD-595. When specified, the exterior finish color shall be tan 686A, Color Number 33446 of SAE AMS-STD-595. The tank boarding ladder shall not be painted.

3.2.1.4.1 Rustproofing. When specified (see 6.1), the HSV cab and chassis shall be rust proofed to a tropical level in accordance with FED-STD-297.

3.2.1.5 Exclusion of water. The design of the HSV shall be such as to prevent water leaking into, or being driven into, any part of the HSV interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the HSV that could collect and retain water.

3.2.1.5.1 Fluid traps and faying surfaces. There shall be no fluid traps on the HSV. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.2.1.5.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

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3.2.1.5.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The minimum size of the drain holes shall be 0.375 inch.

3.2.1.6 Sense line material. All sense lines (both fuel and air) shall be made of Kynar or equivalent materials in order to provide maximum resistance to UV damage and increase the longevity of the vehicle. All lines shall have some form of marking or coloring to indicate whether it is an air, fuel, or another type of sense line.

3.2.2 Markings. The HSV shall be marked for the appropriate service using MIL-HDBK-1223 for guidance. The HSV shall also be marked in accordance with 49 CFR 172.302 and the National Fire Protection Association (NFPA) 407. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch-high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595. The center of gravity of the HSV shall be marked on the unit within 1-inch of the calculated center of gravity. Tire inflation pressure shall be marked as near to each tire as practicable.

3.2.2.1 Marking panel. The markings in the following sections shall be applied to a panel that is locally fabricated from noncorrosive material.

3.2.2.1.1 Flammable. “FLAMMABLE” in 4-inch letters decals or stencils shall be located on each side of the HSV.

3.2.2.1.2 No smoking within 50 feet. “NO SMOKING WITHIN 50 FT” in 3-inch letters shall be located on each side of the HSV.

3.2.2.1.3 Cargo fire avoid water. “CARGO FIRE AVOID WATER” in 2-inch letters shall be located on each side of the HSV.

3.2.2.1.4 Jet fuel product. 3-inch letters shall be located on each side and the rear of the HSV to display the following product markings as described in the table below. The government shall notify the contractor of the appropriate label marking for each HSV ordered (see 6.1).

TABLE I. Jet fuel product markings.

HSV Final Destination Location	Product Markings
CONUS locations except Grand Forks AFB ND, Edwards AFB CA, Beale AFB CA	“Jet Fuel Jet A” and “F-24”
All OCONUS locations and Grand Forks AFB ND, Edwards AFB CA, Beale AFB CA	“Jet Fuel JP-8” and “F-34”

3.2.2.2 Hazardous material decals. Four DOT hazardous material labels, number 1863, shall be mounted on front, rear, left, and right sides of the HSV.

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3.2.2.3 No smoking label. A “NO SMOKING” label shall be attached in the cab in a highly visible location.

3.2.3 Identification and information plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be Securely attached to the HSV in a readily accessible location. The HSV shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. The HSV UID shall be the Vehicle Identification Number (VIN) and shall be marked as both a bar code and human readable markings. The identification plate shall contain the following information:

a. Nomenclature.
b. Air Force vehicle registration number.
c. Part number.
d. Serial number.
e. Date of manufacture.
f. Contract number.
g. Manufacturer’s name.
h. Manufacturer’s address and phone number.
i. Commercial and Government Entity (CAGE) code.
j. Date of warranty expiration.
k. National Stock Number (NSN).
l. Vehicle weight, unloaded (pounds).
m. Gross vehicle weight (pounds).
n. Fuel type.
o. Engine oil viscosities for ambient temperatures ranging from -40°F to +125°F.

3.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the HSV in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the HSV.
b. Horizontal and vertical location of the center of gravity of the HSV in air transportable configuration, marked on the silhouette views.
c. Shipping weight.
d. Loading cubage.
e. Overall height, width, and length.
f. Front and rear axle loads.
g. Tie down information.

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3.2.3.3 Lubrication data plate. The HSV shall be provided with a permanently marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the HSV that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 3.3.1.

3.2.3.4 Control marking. All controls, valves, gauges, and indicators used in the operation of the HSV shall be identified by securely attached nameplates of such construction that exposure to oil, dirt, light, etc., will neither fade nor eradicate them. Tags or decals shall not be used.

3.2.3.5 Operating instructions and data plates. All data plates, placards, charts, instruction plates, etc., shall utilize anodized or coated aluminum plates of a size and shape consistent with the information required thereon.

3.2.3.6 Diagrams. Schematic diagrams of the fuel flow and pneumatic systems shall be provided.

3.2.3.7 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate system operator control panel. The instructions shall be clear, concise, and adequate to enable operation of the HSV without damage to the equipment or injury to personnel and shall refer to the components as identified on the schematic diagrams.

3.2.4 Environment, Safety, and Occupational Health (ESOH).

3.2.4.1 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.2.4.2 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the HSV with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.2.4.3 Fire extinguishers. Two dry chemical fire extinguishers, Type 1, Class 2, Size 20 in accordance with A-A-393.3, shall be mounted with one on each side of the truck. The fire extinguishers shall be mounted in a readily accessible location to personnel standing on the ground and shall be protected from tire splash.

3.2.4.4 Noise. The design shall ensure that noise created by the HSV is compatible with the environment and minimize exposure of personnel to noise hazards during operations and maintenance activities. The maximum A-weighted sound levels produced by the truck during defuel servicing pumping operations shall not exceed 84 dBA at a distance of 15 feet from the geometric center of the truck cab and at the operator’s position in front of the operator control panel.

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3.2.4.5 Electrostatic discharge (ESD). The design of the HSV shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

3.2.4.6 Grounding cables and reels. Two Type I, spring rewind, grounding reels with cables and cable guides shall be installed side by side on the driver's side of the truck. The reels and cables shall be in accordance with A-A-50696. One cable shall have a welder style clamp, and the other cable shall have a grounding plug. The cable reel shall be fitted with a cable guide that allows deployment and rewinding of the cable in a tangle-free manner. The cables shall have cable stops to prevent the clip or plug from striking the cable guide. The cables shall be electrically bonded to the chassis frame.

3.2.4.7 Safety. The truck shall comply with all applicable requirements of the Federal Motor Carrier Safety Regulations (49CFR 393) and Federal Motor Vehicles Safety Standards (49CFR 571) at the time of manufacture. Other than the lift platform (if ordered), the principal walking surfaces shall be of an anti-skid self-cleaning type.

3.2.5 Electromagnetic interference (EMI). The HSV shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 (to the limits of the FIGURE RE102-4, Navy Fixed & Air Force curve) and RS103 (to the levels of TABLE VII) for the following ranges:

a. 30MHz – 1GHz at 10 Volt per meter (V/m).

b. 1GHz – 18GHz at 50V/m.

3.2.6 Cab and chassis.

3.2.6.1 Cab. The R-12 shall be a two door, 2-person capacity cab with two individual air ride seats. All necessary controls, instruments, gauges, and driver's aids to operate the truck shall be located in the cab. The cab shall be air conditioned.

3.2.6.2 Mirrors. The HSV shall be equipped with both left and right-side mirrors. The mirrors shall be electronically adjustable from the driver's position within the cab. The mirrors shall be large enough to provide a clear and distinct view of the entire length of the truck and rear bumper.

3.2.6.3 Backup aids. The HSV shall be equipped with backup lights and a backup alarm. The alarm shall meet the Type D (87dbA) requirements of SAE J994.

3.2.6.4 Chassis. The truck chassis shall be a standard commercial chassis in accordance with FED-STD-794 for a four by two chassis with a two-door cab, Type 1, Class D, as modified to comply with the specific requirements herein. The R-12 shall be towable from the front with or without the front tires off the ground.

3.2.6.5 Lighting. Unless lights are provided as standard equipment for the commercial chassis, all lighting for the HSV shall be LED except for headlights.

3.2.6.6 Equipment and accessories. In addition to the cab requirements of FED-STD- 794 for equipment and accessories, the R-12 shall be equipped with an hour meter and two switched reading lights for use while seated or while standing at either door during night operations.

3.2.6.7 Wheel chock storage container. One commercial wheel chock and a container for the chock shall be provided on the left rear of the truck behind the axle.

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3.2.6.8 Fenders. The HSV shall have metal fenders or similar protective covers over the wheels. The fenders shall support a 250-pound person at any point that can be stepped on without permanent deformation.

3.2.6.9 Mud flaps. Mud flaps shall be installed at the rear of each fender.

3.2.7 Pumping system. The HSV shall have both a flow control system and a pressure control system that consists of the necessary control valves, pilot valves, selector controls, automatic controls, venturis, gauges, surge suppression devices, and safety devices necessary to regulate the flow and pressure of fuel during refueling and defueling on Type III hydrant systems (minimum pressure of Type III hydrant system is 120 psi while maximum pressure is 160 psi).

3.2.7.1 Flow control system. The flow control system shall control flow through the ground discharge hoses and through the lift platform hoses (if ordered, see 5.1) by initiating and stopping flow with the dead man control system. The flow control system shall not allow the flow rate to exceed the rated flow of the filtration system.

3.2.7.1.1 Flow control valve. The flow control valve shall have an opening rate adjustable from 2 to 25 seconds and a closing rate adjustable from 10 to 30 seconds.

3.2.7.2 Pressure control system. The pressure control system shall sense fuel pressure through some form of pressure compensator in order to simulate single-point nozzle pressure at the aircraft. The HSV shall provide the sense pressure to a Type III hydrant system pit valve. The Type III pit valve serves as the primary single point nozzle pressure control and is typically set at 40 +/- 5 psi to limit the single point nozzle pressure at the aircraft. An independent secondary single point nozzle pressure control shall also be installed on the HSV and will limit the single point nozzle pressure at the aircraft to 50 +/- 5 psi should the primary pressure control fail.

3.2.7.2.1 Pressure surges. Pressure surges due to a quick two second shutdown at the aircraft shall not exceed the published rating of any component in the fuel flow system.

3.2.7.2.2 Pressure relief system. A pressure relief system shall be provided to relieve pressure in the pumping system after fueling operations are complete. The pressure relief system shall not allow backflow of fuel into the recovery tank (see 3.2.7.2.3).

3.2.7.2.3 Fuel recovery tank. A fuel recovery tank with a minimum 30-gallon capacity shall be provided to allow for pressure relief of the pumping system during fueling operations, defueling operations, thermal expansion of fuel, and fuel from the fuel filtration system air eliminator. The tank shall be equipped with relief valves and vents as necessary for safe operation. A gauge shall be provided on the tank to monitor the level of fuel and shall be readable by an operator standing on the ground.

3.2.7.2.3.1 Fuel recovery tank drain. The fuel recovery tank shall be equipped with a manual drain valve to allow for complete draining of the tank.

3.2.7.2.3.2 Automatic emptying system. The fuel recovery tank shall have an automatic tank emptying system to empty the fuel recovery tank during both refueling and defueling operations and shall incorporate a full tank shutdown system. The automatic emptying system shall not utilize an electrically driven pump and shall not allow the backflow of fuel into the recovery tank when not in use. The automatic emptying system shall return the fuel up-stream of the fuel filtration system. The automatic emptying system shall only operate when the dead man control is activated.

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The system shall also incorporate an override momentary-on push button located on the operator control panel to allow the manual operation of the automatic emptying system.

3.2.7.2.3.3 Warning light. A red high-level warning light shall be installed on the operator control panel. This light shall illuminate whenever the fuel level in the recovery tank reaches the predetermined level where the automatic return system is scheduled to activate. The high-level recovery tank warning light shall remain illuminated until the fuel level in the recovery tank drops below the system activation point.

3.2.7.3 Defuel system. The HSV shall be provided with a defuel system capable of defueling aircraft through either the ground servicing hoses, or the lift platform hoses (if lift platform is ordered, see 5.1) into a Type III hydrant system. The defuel system shall be capable of defueling at a minimum flow rate of 80 GPM through one rear ground discharge hose without the assistance of aircraft boost pumps and 300 GPM with the assistance of aircraft boost pumps (aircraft boost pumps are capable of providing 50 psi at the single point nozzle). The defuel suction line shall incorporate a, 8 to 14 mesh low pressure drop strainer that is easily removed for cleaning. The defuel system shall be activated through a defuel mode selector switch located on the operator control panel.

3.2.7.4 Piping. The piping shall be seamless schedule 40 aluminum in accordance with ASTM B241/B241M or seamless schedule 10 TP 304L stainless steel in accordance with ASTM A312/A312M. Pipe and fittings shall be either flange or groove connected, or a combination. Coupling grooves shall be cut or rolled in accordance with coupling manufacturer's requirements and recommendations. Fuel pressure build-up in the piping and hose(s) shall relieve into the fuel recovery tank. Pipe mounting shall prevent failure due to chaffing, vibration, or movement, due to operational or mobility induced forces. Piping shall be protected when passing through sheet metal and shall not be used as a step.

3.2.7.5 Volumetric flow meter. All fuel flow through the pumping system shall be metered through the flow meter. The meter shall be an electronic, positive displacement flow meter that is certified accurate to ± 0.2 percent between 50 and 1000 GPM in accordance with NIST Handbook 44. The meter shall have an LED digital display and be capable of displaying totalized flow and flow rate in gallons per minute. The meter shall have both a resettable counter and a non-resettable totalizer. The meter shall be readable from a 15-foot distance during both daytime and nighttime operations.

3.2.8 Filtration system. The filtration system shall be rated for the maximum flow and pressure of the HSV. The filter vessel and elements shall be qualified in accordance with the requirements of EI 1581, Specification and Qualification Procedures for Aviation Jet Fuel Filter Separators, Category M, Type S, Two Stage System Configuration.

3.2.8.1 Filter vessel. The filter vessel shall be designed and constructed in accordance with the most recent edition and revision of ASME Boiler and Pressure Vessel Code Section VIII. The vessel shall be inspected and stamped by a qualified ASME inspector. The filter vessel shall be constructed of aluminum and shall include a ball valve vent to aid in draining of the filter vessel. The ball valve vent shall include a check valve for only allowing air in. The filter vessel shall remain full regardless of the level of fuel in the cargo tank during pumping operation or while taking a sample from the drain. The filter vessel drain shall have a self-closing valve, accessible by a person standing on the ground.

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3.2.8.2 Air eliminator. An air eliminator shall be provided in the top of the filter vessel to allow automatic purging of air from the pumping system. The eliminator shall be vented into the fuel recovery tank.

3.2.8.3 Water-slug shutoff device. A water-slug shutoff device that uses a float-actuated valve or an electronic sensor shall be installed in the filter vessel sump. The device shall stop fuel flow within 10 seconds when a predetermined water level is reached inside the sump. If using a float-actuated valve, the valve shall include provisions that permit manual movement of the float through its complete range of motion for testing purposes when the system is pressurized. The manual test feature shall be accessible by an operator standing on the ground. If an electronic sensor is used, it must have a light or other visual means to show proper working order. The electronic sensor must also be certified in accordance with EI 1592.

3.2.8.4 Filter vessel drain. The water sump of the filter vessel shall be equipped with a one inch manually operable ball-type valve with a spring return handle that renders the valve normally closed. The drain valve shall be: (1) designed to prevent fuel contact with the operator, (2) directed toward the ground, (3) guarded from damage, and (4) accessible without crawling under the HSV.

3.2.8.5 Element sealing. Element connections shall be sealed in accordance with EI 1581.

3.2.9 Hoses, reels, and nozzles.

3.2.9.1 Ground servicing hoses. Unless otherwise specified, the HSV shall be equipped with two 60-foot ground servicing hoses. The hoses shall have a 2-1/2” inside diameter and conform to EI 1529, Grade 2, Type C hose specifications. When specified (see 6.1) for extreme climatic temperature locations, the hoses shall have a 2-1/2" inside diameter and conform to EI 1529, Grade 2, Type CT hose specifications. All hoses shall be equipped with couplings in accordance with EI 1529. The couplings shall be two-piece construction, internally expandable, non-reattach able design, constructed of high strength corrosion resistant material.

3.2.9.2 Ground servicing hose reel. The ground servicing hose reels shall be air powered or electric powered commercial hose reels located at the rear of the vehicle. The reels shall have a clutch released drag brake, a manual rewind, and a locking mechanism that will ensure the hose does not come loose during transportation. The locking mechanism shall be easily accessible by an operator standing on the ground. The hose reel shall be capable of holding 75 working feet (from the hose roller guide to the end of the single point nozzle) of 2.5-inch non-collapsible hoses with 2.5-inch National Pipe Threat Tapered Thread (NPT) male threads on each end. A momentary-on push button for rewinding the hose shall be provided in an easily accessible location adjacent to the reel. Unless the hose reel is of the single wrap type, stainless steel hose guide rollers shall be used. The rollers shall have permanently lubricated metal bearings. A manual isolation valve shall be provided for the hose reel. The hose reels shall have provisions to prevent over-travel during deployment and rewind. The crank required for manual rewind shall be stored on the HSV. All moving parts (drive chains, gears, etc.) shall be equipped with protective coverings to prevent injury to the operator or damage to the hoses.

3.2.9.3 Lift platform hoses. If the lift platform is ordered (see 5.1), the lift platform shall be equipped with two ground servicing hoses that are long enough to connect to a single point receptacle five feet on either side of the truck center line. Unless otherwise specified, the hoses shall have a 2-1/2” inside diameter and conform to EI 1529, Grade 2, Type C hose specifications. When specified (see 5.1), the hoses shall have a 2-1/2” inside diameter and conform to EI 1529,

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Grade 2, Type CT hose specifications. All hoses shall be equipped with couplings in accordance with EI 1529. The couplings shall be two-piece construction, internally expandable, non-reattach able design, constructed of high strength corrosion resistant material. Provisions shall be made for storage of the hose on the lift platform, so the hose will not be chaffed, twisted, crimped, or pinched under any operating condition.

3.2.9.4 Hydrant hose. The HSV shall be equipped with a 30-foot-long hydrant hose. Unless otherwise specified, the hose shall have a 4" inside diameter and conform to EI 1529, Grade 2, Type C hose specifications. When specified (see 6.1), the hose shall have a 4" inside diameter and conform to EI 1529, Grade 2, Type CT hose specifications. The hose shall be equipped with couplings in accordance with EI 1529. The couplings shall be two-piece construction, internally expandable, non-reattach able design, constructed of high strength corrosion resistant material.

3.2.9.5 Hydrant hose reel. The hydrant hose reel shall be an air powered or electric powered commercial hose reel locate on the driver's side of the truck. The reel shall have a clutch released drag brake, a manual rewind, and a locking mechanism that will ensure the hose does not come loose during transportation. The locking mechanism shall be easily accessible by an operator standing on the ground. The hose reel shall be capable of holding 30 working feet (from the hose roller guide to the end of the hydrant coupler) of 4-inch non-collapsible hoses with 4-inch National Pipe Threat Tapered Thread (NPT) male threads on each end. A momentary-on push button for rewinding the hose shall be provided in an easily accessible location adjacent to the reel. Unless the hose reel is of the single wrap type, stainless steel hose guide rollers shall be used. The rollers shall have permanently lubricated metal bearings. A manual isolation valve shall be provided for the hose reel. The hose reel shall have provisions to prevent over-travel during deployment and rewind. The crank required for manual rewind shall be stored on the HSV. All moving parts (drive chains, gears, etc.) shall be equipped with protective coverings to prevent injury to the operator or damage to the hoses. The hydrant coupler shall be no higher than 3 feet above the ground when in the stowed position.

3.2.9.6 Hydrant sensing lines. Sensing lines for connecting to the Type III hydrant system shall be provided. Two lines will be required for the connection to the hydrant pit valve: one for air pressure, and one for fuel pressure. The lines shall not exceed the length of the hydrant hose. Dummy sensing line connections shall be equipped adjacent to the sensing line reel to secure the lines when not in use.

3.2.9.7 Hydrant sensing line reel. The hydrant sensing line reel shall be of the spring rewind type and shall be located next to the hydrant hose reel. The reel shall be equipped with arresting features for making random stops at any position. Provisions shall be made to prevent the last half turn of hose from unwinding, and a ball stop shall be provided to prevent over wind.

3.2.9.8 Single point nozzles. Both ground servicing hoses shall be equipped with a type D3 single point nozzles in accordance with SAE- AS-5877. If the lift platform is ordered (see 5.1), both lift platform hoses shall be equipped with a type D2 single point nozzles in accordance with SAE-AS-5877. All nozzles shall be equipped with a ball valve strainer (Cla-Val P/N E349GF or JC Carter P/N 64015), have a 60-mesh stainless steel strainer, and an automatic vacuum breaker. All nozzles shall have stowage receptacles with interlocks that prevent the parking brake from releasing unless the nozzles are properly stowed. The stowage receptacles and interlocks shall be compatible with all single point nozzles complying with the applicable version or SAE-AS-5877.

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3.2.9.9 Hydrant coupler. Unless otherwise specified the hydrant hose shall be equipped with a 4-inch API style coupler (Cla-Val 351GF-15 or equivalent) in accordance with EI 1584 for use with Type II and III hydrant systems. When specified (see 6.1), the hydrant hose shall be equipped with a 4-inch DOD style coupler (Cla-Val 351GF-14 or equivalent) for use with Type I hydrant systems. The nozzle shall have a stowage receptacle with an interlock that prevents the parking brake from releasing unless the nozzle is properly stowed.

3.2.10 Dead man control. An electric or pneumatic dead man controller in accordance with NFPA 407 shall be provided to start and stop flow to and from the aircraft. The dead man control shall be a robust squeeze handle or push-button type device suitable for refueler service. When the dead man is released, the flow of fuel shall be stopped by closing of the hydrant pit valve and by closing of a valve downstream of the hydrant coupler. If an inline control valve is provided, it shall be utilized as the valve downstream of the hydrant coupler. The dead man cable or hose shall have a length of 70 feet and be positioned so that the dead man control can be activated from the operator control panel, the ground discharge hose reels, and the raised lift platform (if ordered, see 6.1). If an electric dead man is used, a cable reel with spring rewind shall be provided. If a pneumatic dead man is used, a powered cable reel with a momentary-on push button shall be used for rewinding the hose. In either case, a cable or hose stop to prevent the dead man device from striking the reel guide shall be provided, and the reel shall be conveniently located near the operator control panel.

3.2.11 Operator control panel. A covered operator control panel shall be located on the driver's side of the HSV adjacent to the hydrant hose reel. The operator control panel shall be fully illuminated for nighttime servicing operations. The operator control panel shall be easily accessible for maintenance from the rear and may swing out, if necessary. Instructions and diagram plates shall be mounted on or adjacent to the operator control panel. All components on the panel shall be weatherproof. The operator control panel controls shall include but not be limited to:

a.) Defuel mode selector (see 3.2.7.3).

b.) Manual recovery tank override button (see 3.2.7.2.3.2).

c.) Recovery tank high-level warning light (see 3.2.7.2.3.3).

d.) Auxiliary throttle control (see 3.2.11.1).

e.) Tachometer (see 3.2.11.2).

f.) Emergency engine shut-off switch (see 3.2.11.3).

g.) Emergency fuel shutoff (see 3.2.11.4).

h.) Pressure gauges (see 3.2.11.5).

i.) Differential pressure gauge (see 3.2.11.6).

j.) Surge suppression device fill connections if required (see 3.2.11.7).

3.2.11.1 Auxiliary throttle. An auxiliary throttle, adjustable to within 50 revolutions per minute (rpm) of the maximum rated engine speed or maximum pump speed, shall be provided and utilized during defuel servicing operations. The auxiliary throttle shall be used for controlling flow and shall be located on the operator control panel. The auxiliary throttle shall hold any speed setting and provide an emergency return to idle feature.

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3.2.11.2 Tachometer. An electronic, LED digital display, tachometer shall be provided. The tachometer shall incorporate numerical value readout and shall be readable from a distance of 15 feet, during daytime or nighttime operations.

3.2.11.3 Emergency engine shutoff. An emergency engine shut-off switch shall be located on the operator control panel and shall shut down the truck engine without the use of any other control. The switch shall be a push-pull mushroom type switch.

3.2.11.4 Emergency fuel shutoff. The HSV shall incorporate two (three if equipped with a lift platform) independent emergency fuel shut-off control switches. The emergency fuel shut-off control switches shall be located as follows: (1) one on the operator control panel, (2) one on the passenger's side of the HSV opposite the operator control panel, and (3) one on the gauge panel of the lift platform if provided. The emergency fuel shut-off control shall be a toggle switch that is either activated or push-pull. If a push-pull type of air operated valve is used, then the valve shall be a two-position detent valve which is easily operated and arranged so that it is pushed to activate. The emergency fuel shut-off controls shall immediately terminate the fuel flow in any mode regardless of whether the deadman device is activated or deactivated. The emergency fuel shut-off control shall not be recessed.

3.2.11.5 Pressure gauges. Electronic, LED digital display, pressure gauges shall be installed on the operator control panel to display the pressure readings for pump discharge pressure, hydrant inlet pressure, and single point nozzle pressure. The pressure readings shall be taken from pressure transmitters that are incorporated into the flow and pressure control system. All displays shall be readable from a distance of 15 feet during daytime or nighttime operations.

3.2.11.6 Differential pressure gauge. A non-calibrating differential pressure gauge shall be installed on the operator control panel to continuously monitor the pressure drop across the filter coalescer stage during servicing operations and shall register and hold the highest pressure drop achieved during servicing operations.

3.2.11.7 Surge suppression device fill connections. The HSV shall incorporate surge suppression devices for controlling surge. If the devices require maintaining a specific internal pressure and must be serviced periodically, the fill connections and associated pressure gauges for servicing the devices shall be located on the operator control panel.

3.2.12 Sampling devices. The main flow system shall be equipped with two sampling connections: one upstream of the filter separator near the hydrant hose outlet, the other downstream of the filter near the discharge hoses. The fuel sampling devices shall be easily accessible while standing on the ground. Adequate space shall be provided to connect an in-line sampler such as the Millipore Corporation, Bedford, MA 01730, fluid sampling kit, catalog No. XX64 037 00, for solid and water samples. The sampling device shall consist of the necessary corrosion resistant piping, a one quarter turn ball valve, and a quick disconnect with a dry break coupler and dust plug for connection to the sampling kit. In addition, a Gammon Mini Monitor Tray Assembly Model No. GTP-9589 or equivalent shall be provided and shall be easily accessible while standing at the fuel sampling device near the control panel area. A three-way selector valve is not required. The Gammon Mini Monitor shall be plumbed so that it samples fuel on the downstream side of the filter separator.

3.2.13 Lift platform. When a Type I HSV is specified (see 5.1), the HSV shall be provided with a variable height lift platform that has a minimum lift capacity of 2000 pounds. The lift platform

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deck shall not exceed 58 inches from the ground when in the stowed position and shall be located adjacent to the rear of the cab. The lift platform shall raise from the stowed position to the height required to allow a small human, as defined by MIL-STD-1472, to service any wide-bodied aircraft at the under-wing receptacle. The lift platform shall be capable of being lowered to the lowest position with the lift platform hoses still connected to the aircraft. The platform shall be capable of being fully raised or lowered in not less than 15 seconds or more than 35 seconds. A means shall be provided to prevent rapid accidental lowering of the platform in the event of hose or other component failure. A lowering device (manual override) that will override the main control and provide a smooth controlled lowering of the platform shall be provided in a convenient location accessible by the operator from the ground. The lift platform shall have a wing stop control which will stop the platform from lifting before the highest portion of the lift comes in contact with the aircraft wing. The lift platform system shall incorporate a safety interlock that sets the disables the lift system unless the parking brakes are set and does not allow the parking brake to be released when the lift platform is not in the full down position. The lift platform shall meet all requirement of 29CFR 1910.67 and shall be equipped with the following features:

a. Aluminum guard rails with an entry gate, an open grate deck, and a toe plate.

b. An access ladder or steps.

c. single-point nozzle pressure gauge.

d. Lift controls.

e. An emergency engine shut-off switch.

f. An appropriately sized, weather protected, operator control panel incorporating the necessary controls for proper operation of the lift platform.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The HSV shall be capable of operating in ambient temperatures ranging -40° F to 125° F.

3.3.2 Storage temperature range. The HSV shall be capable of being stored in ambient temperatures ranging from -40° F to 125° F.

3.3.3 Solar radiation. The HSV shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

3.3.4 Fungus. All materials used in the HSV shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The HSV shall be suitable for operation and storage in conditions encountered in a tropical environment.

3.3.5 Salt fog. The HSV shall be capable of storage and operation in high temperature, high humidity, salt laden, seacoast environments without damage or deterioration of performance.

3.3.6 Sand and dust. The HSV shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

3.3.7 Humidity. The HSV shall be certified capable of storage and operation at relative humidity from 0 to 100 percent.

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3.4 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 3.5.2) shall not exceed:

a. Weight 26,000 pounds.

b. Length 320 inches.

c. Width 96 inches.

d. Height 102 inches.

3.5 Transportability.

3.5.1 Surface transportability. The HSV shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

3.5.2 Air transportability. The HSV shall be transportable on C-130, C-5, and C-17 aircraft in accordance with MIL-STD-1791. In all air transport configurations, the HSV shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The HSV shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The HSV shall drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.5.2.1 Shoring. If shoring is necessary for loading the HSV on a C-130, C-17, or C-5 aircraft, the shoring shall be supplied with each Refueling Vehicle.

3.5.2.2 Axle weight. Axle weight shall not exceed 13,000 pounds.

3.5.2.3 Tire pressure. Tire pressure shall not exceed 100 psi and shall not be reduced for air transport.

3.5.2.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 15 minutes and restoration to operating configuration shall take no more than 15 minutes for two persons using common hand tools (see 7.2.1). All equipment removed shall be stored on the HSV; caps and plugs shall permit moving and storage in transport configuration.

3.5.3 Tie downs. The HSV shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the HSV. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.6 Maintainability. All assemblies, controls, and installed equipment shall be located so that there is no interference with each other or the operation, and shall be readily accessible for maintenance operation, removal, and replacement using common hand tools (see 7.2.1). All operating controls and servicing shall be so designed to allow access by personnel wearing arctic clothing and mittens/gloves.

3.6.1 Inspection and servicing provisions.

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a. Routine servicing tasks and pre-use inspections shall require no hand tools.

b. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

c. The HSV shall be designed with maximum usage of sealed lifetime lubrication bearings.

d. The HSV shall be designed so the correct oil and coolant levels can be visually checked while the unit is running.

3.6.2 Special tools. The design of the HSV shall minimize the requirement for special tools. All special tools (see 7.2.2) shall be provided and stored on the HSV.

3.6.3 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the HSV shall be provided with each HSV on CDROM or DVDROM.

3.7 Mobility. The fully equipped HSV loaded to its gross vehicle weight rating (GVWR) shall:

a. Speed. Maintain a speed of 55 miles per hour (mph) on a dry level paved road and 40 mph over an unimproved road, without damage to the chassis or any components.
b. Acceleration. Accelerate from 0 to 15-mph within 60 seconds on a dry level paved road.
c. Clearances. The fully loaded HSV shall be capable of negotiating an eight-inch-high radius top berm, with 15 percent approach and departure angles (simulating a secondary containment barrier around a fill stand) in a straight pass as well as a minimum radius turn mid-way in either direction. There shall be no structural failure of any component, including permanent deformation, or evidence of impending failure. There shall be no interference between the HSV components or between the HSV, the ground, and all required obstacles, except for normal contact by the tires.
d. Turning diameter. Have a wall to wall turning diameter, as defined by SAE-J695, of 99-feet maximum in both directions.

3.7.1 Transmission. The HSV shall be equipped with an automatic transmission which shall include a hydraulic torque converter and not less than five forward gear ratios. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

3.7.1.1 Power take-off (PTO). The defuel system pump shall be driven via a transmission driven PTO or split shaft gearbox design.

3.7.2 Axles. The HSV shall be equipped with axles rated at least equal to the load imposed on each axle, measured at the ground, with the HSV loaded to its GVWR.

3.7.3 Suspension. The HSV shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the HSV loaded to its GVWR. The HSV shall have a rear air suspension system.

3.7.4 Steering. The HSV shall be equipped with a power assisted steering system, with manual steering available in the event of power assist failure.

3.7.5 Brakes. The HSV shall be equipped with an air brake system.

3.7.6 Tires and wheels. The HSV shall be equipped with commercially available, identical, wheels and tires and shall be in accordance with the Tire and Rim Association requirements for this application.

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3.7.6.1 Spare tire and wheel assembly. When specified (see 6.1), a spare tire and wheel assembly (the tire mounted on the wheel) shall be mounted on the HSV; the assembly shall be identical to the others provided. A mount point on the HSV is not required. The spare tire and wheel assembly shall be secured in the lift platform (if ordered) or in the passenger side of the cab, without damaging any components or material.

3.7.6.2 Spare wheel. When specified (see 6.1), a spare wheel shall be mounted on the HSV; the wheel shall be identical to the others provided. A mount point on the HSV is not required. The wheel shall be secured in the lift platform (if required) or in the passenger side of the cab, without damaging any components or material.

3.8 Engine and related equipment. The HSV shall be equipped with Environmental Protection Agency Tier III diesel engine. This requires a National Security Exemption (NSE) from the Clean Air Act. The engine shall be certified to comply with the most stringent Environmental Protection Agency (EPA) non-road diesel engine emission requirements possible while meeting the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur.

3.8.1 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake. The exhaust system shall meet all applicable requirements of the most recent edition and revision of NFPA 407.

3.8.2 Engine fuels and fuel system.

3.8.2.1 Engine primary fuels. The following shall be primary fuels (see 7.2.6):

a. JP-5, in accordance with MIL-DTL-5624, -40° F to 125° F ambient air temperature.

b. JP-8 in accordance with MIL-DTL-83133 -40° F to 125° F ambient air temperature.

c. F-24, in accordance with NATO AFLP-3747, -40° F to 125° F ambient air temperature.

3.8.2.2 Engine alternate fuels. The following shall be alternate fuels (see 7.2.5):

a. 1-D S15, 1-D S500, 1-D S5000 in accordance with ASTM D975, below 60° F ambient air temperature.

b. 2-D S15, 2-D S500, 2-D S5000 in accordance with ASTM D975, above 32° F ambient air temperature.

c. TS-1 in accordance with GOST 10227-86 with JP-8 additives -40 ° F to 125° F ambient air temperature.

3.8.2.3 Engine emergency fuels. The following shall be emergency fuels: Jet A, Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

3.8.3 Engine operator instruments. The following instruments shall be provided to the engine operator:

a. Tachometer.

b. Coolant temperature gauge.

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c. Low coolant level indicator.

d. Oil pressure gauge.

e. Oil temperature gauge.

f. Fuel level gauge.

g. Hour meter.

3.9 Electrical system. The HSV wiring system and electrical components shall be in accordance with NFPA 407 and SAE ARP 1247. All wiring rear of the cab shall be run in plastic/metallic tubing and shall utilize vapor proof connectors. All electrical enclosures shall be vapor proof. All exposed connections shall be weather protected.

3.9.1 Batteries and battery compartment.

3.9.1.1 Batteries. Batteries shall be maintenance free and easily accessible for daily inspection without the need for hand tools. Batteries shall be commercially available and shall be of the top post type.

3.9.1.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

3.9.1.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings and shall not be spliced.

3.9.1.4 Jump Start Connection. The Refueling Vehicle shall be equipped with an external power receptacle for an SAE-AS90347 type plug to facilitate jump starting operations.

3.9.2 Winterization system. A winterization system shall be provided for starting in temperatures below 0° F. The winterization system shall include heaters for engine coolant, engine oil, and the fuel tank, as well as battery warmers. The winterization system shall be designed to operate from both an external 120/208-volt AC, 60 Hz power source utilizing the external electrical connections and an external 220/308-volt AC, 50 Hz power source utilizing external connections. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. A labeled indicator light shall be installed on the instrument panel to indicate when the vehicle is connected to external power.

3.9.2.1 Additional moisture removing device. When specified (see 6.1), an additional moisture removing device for the truck system air shall be included.

3.9.3 Plugs and connectors. Plugs and connectors shall be in accordance with NEMA WD 6.

3.9.4 Floodlight. A fully adjustable floodlight shall be provided on the forward left side of the HSV. The light shall illuminate the entire left side of the HSV. A fully adjustable floodlight shall be provided on the rear of the HSV. The rear light shall illuminate the entire rear working area of the HSV. An individual switch and adjusting handle shall be provided for each light to allow operation from the ground. The lights shall not operate unless the HSV's parking lights are on. The lights shall have a minimum of 1000 candlepower.

3.10 Technical data.

3.10.1 Manufacturer publications. The contractor shall provide the following manufacturer's publications in Indexed Portable Document Format (IPDF). One digital copy on disc shall be

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furnished to AFLCMC/WNZ 30 days prior to the first hardware delivery. All manuals shall be delivered with either unlimited data rights or as a minimum, a Government Purpose License, to allow the Air Force to electronically store, print, distribute and update the manual. Manuals are subject to government review and approval in accordance with MIL-PRF-32216. Any changes or corrections noted by the Air Force shall be corrected and updated pages or manuals shall be submitted. Only one digital copy is required per order regardless of quantity of units.

3.10.1.1 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the vehicle. The operator's manual shall include at least the following:

a. General description and step-by-step instructions for the operation of the vehicle and its auxiliary equipment.

b. Checklists, lubrication charts, and basic troubleshooting procedures for the daily maintenance inspection and mission readiness checks that the operator is expected to perform.

c. Schedules for required preventative maintenance and required periodic maintenance.

d. Full descriptions and illustrations for the location and function of all controls.

e. List of all checks and adjustments needed in preparation for placing the vehicle into service upon receipt from the contractor.

f. Preparation Instructions for shipment or storage.

g. Warranty information and a list that details the period of the warranty for the complete vehicle and for any component that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers listed.

3.10.1.2 Service and repair manuals. Service and repair manuals shall be provided for the chassis, engine, transmission, tanker body, and equipment. The service manuals shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, fluid capacities, current data, voltage data, resistance data, test procedures, and any illustrations or exploded views that may be required to perform proper maintenance by qualified mechanics. The service and repair manuals shall contain an alphabetical subject index as well as a table of contents. The manual shall contain at least the following where applicable:

a. Pneumatic, electrical, and fuel schematics.

b. Schedules for required preventative maintenance and required periodic maintenance.

c. Troubleshooting, servicing, disassembly, repair, and assembly procedures.

3.10.1.3 Illustrated parts breakdown. An illustrated parts breakdown (IPB) manual shall be provided for the chassis, engine, transmission, tanker body, and equipment. IPB manuals shall include illustrations and exploded views as needed to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts list. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The IPB manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics

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shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The IPB manuals shall contain a numerical index. The manuals shall contain a list of all the component vendor names, addresses, and telephone numbers referenced in the parts list.

3.10.1.4 Mailing address. The electronic copy of the manufacturer's publications shall be sent to the following address:

a. If sending via USPS:	If sending via Fed Ex or UPS.
b. Support Equipment & Vehicles Division	Support Equipment & Vehicles Division.
c. Refueler IPT	Refueler IPT.
d. AFLCMC/WN-SEV	AFLCMC/WN-SEV.
e. Attn: Mr. Brandon Hammontree	Attn: Mr. Brandon Hammontree.
f. 235 Byron Street, Suite 19A	235 Byron Street, Suite 19A.
g. Robins AFB, GA 31098-1813	Bldg. 300WW South End.

3.10.2 Service bulletins. The contractor shall prepare service bulletins as required to notify the government of new maintenance problems and/or procedures to be implemented on the contractor's equipment prior to the issuance of permanent changes to the technical manuals. Service bulletins shall also be prepared when the program office has identified a trend of a potential problem or failure that has occurred at field level. The service bulletins shall identify any potential safety hazards and shall be submitted in a Microsoft Office digital format.

3.11 Workmanship. The Refueling Vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.11.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be de-burred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.11.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.11.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.11.4 Cleaning. The Refueling Vehicle shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered materials. Recycled, recovered, or environmentally preferable materials (see 7.2.3) should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to

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the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable time frame; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production HSV shall be subjected to the analyses, demonstrations, examinations, and tests described in 5.5 - 5.5.20 The contractor shall provide or arrange for all test equipment and facilities. Testing and demonstrations shall be performed on a simulated Type III Hydrant System (see .7.2.4). Unless otherwise specified, all testing and demonstrations in which the engine is operated shall be performed using JP-8 turbine fuel.

5.3 Conformance inspection. Each production HSV shall be subjected to the demonstrations, and tests described in 5.5.7 - 5.5.20.

5.4 Inspection requirements.

5.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

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5.4.3 Test rejection criteria. Throughout all tests specified herein, the HSV shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including lubricant or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the HSV components or between the HSV, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Evidence of corrosion or deterioration.

5.5 Test methods.

5.5.1 Examination of product. Each HSV shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 5.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each HSV function shall be verified. Certifications and analyses shall be provided in accordance with Table II. Each production HSV shall be inspected to a Government-approved reduced version of the checklist.

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TABLE II. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.2 Design and construction.	Contractor certification that the HSV is designed and constructed in accordance with 3.2.
3.2.1 Materials, protective coatings, and finish.	Contractor certification that the HSV's materials, protective coatings, and finishes are in accordance with 3.2.1.
3.2.1.1 Metals.	Contractor certifications that all metals used in the design of the HSV are in accordance with 3.2.1.1.
3.2.1.2 Elastomers	Contractor certifications that all elastomeric materials used in the design of the HSV are in accordance with 3.2.1.2.
3.2.1.3 Protective coatings.	Contractor certifications that the protective coatings used on materials in the design of the HSV are in accordance with 3.2.1.3.
3.2.1.4 Finish.	Contractor certification that the finish requirements are in accordance with 3.2.1.4.
3.2.1.4.1 Rust proofing.	Contractor certification that rust proofing requirements are in accordance with 3.2.1.4.1.
3.2.1.5.1 Fluid traps and faying surfaces.	Contractor certification that the fluid traps and faying surfaces requirements are in accordance with 3.2.1.5.1.
3.2.1.5.2 Ventilation.	Contractor certification that the ventilation requirements are in accordance with 3.2.1.5.2
3.2.1.5.3 Drainage	Contractor certification that the drainage requirements are in accordance with 3.2.1.5.3.
3.2.1.6 Sense line material.	Contractor certification that the sense line material used is in accordance with 3.2.1.6.
3.2.4.7 Safety.	Contractor certification that the safety requirements are in accordance with 3.2.4.7
3.2.5 Electromagnetic interference (EMI).	Contractor certification that EMI requirements are in accordance with 3.2.5.
3.2.6.4 Chassis.	Contractor certification that the chassis requirements are in accordance with 3.2.6.4.
3.2.8 Filtration system.	Contractor certification the filtration system requirements are in accordance with 3.2.8
3.2.8.1 Filter vessel.	Contractor certification the filter vessel requirements are in accordance with 3.2.8.1.
3.2.8.5 Element sealing.	Contractor certification the filtration element sealing requirements are in accordance with 3.2.8.5.

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TABLE II. Certifications and analyses - Continued

Paragraph	Required Certifications and Analyses
3.2.9.1 Ground servicing hoses.	Contractor certification the ground servicing hose requirements are in accordance with 3.2.9.1.
3.3.1 Operating temperature range.	Contractor certification that the HSV shall be capable of operating in ambient temperatures ranging from -40 ºF to 125 ºF.
3.3.2 Storage temperature range.	Contractor certification that the HSV shall be capable of being stored in ambient temperatures ranging from -40 ºF to 125 ºF
3.3.3 Solar radiation.	Contractor certification that the HSV shall be capable of meeting solar radiation requirements in 3.3.3.
3.3.4 Fungus	Contractor certification that the HSV shall be capable of meeting fungus requirements in 3.3.4.
3.3.5 Salt fog.	Contractor certification that the HSV shall be capable of storage and operation in high temperature, high humidity, salt laden, and sea coast environments without damage or deterioration of performance.
3.3.6 Sand and dust.	Contractor certification that the HSV shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.
3.3.7 Humidity.	Contractor certification that the HSV shall be capable of storage and operation at relative humidity from 0 to 100 percent.
3.5.2 Air transportability.	Contractor certification that air transportability requirements are in accordance with 3.5.2.
3.5.2.1 Shoring.	Contractor certification that shoring shall not be required in accordance with 3.5.2.1.
3.5.3 Tie downs.	Contractor certification that tie downs are in accordance with 3.5.3.
3.6.2 Special tools.	Contractor certification that all special tool requirements are in accordance with 3.6.2.
3.8 Engine and related equipment.	Contractor certification that the engine and related equipment requirements are in accordance with 3.8.

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TABLE II. Certifications and analyses - Continued

Paragraph	Required Certifications and Analyses
3.8.2 Engine fuels and fuel system.	Contractor certification that the HSV engine shall operate on the fuels specified in 3.8.2.
3.9.1.1 Batteries.	Contractor certification that the battery requirements are in accordance with 3.9.1.1.
3.9.1.3 Battery cables.	Contractor certification that the battery cable requirements are in accordance with 3.9.1.3.
3.11 Workmanship.	Contractor certification that workmanship requirements are in accordance with 3.11.
3.11.1 Bolted connections.	Contractor certification that bolted connections requirements are in accordance with 3.11.1.
3.11.2 Riveted connections.	Contractor certification that riveted connections requirements are in accordance with 3.11.2.
3.11.3 Gear and lever assemblies.	Contractor certification that gear and lever assemblies requirements are in accordance with 3.11.3.
3.11.4 Cleaning.	Contractor certification that cleaning requirements are in accordance with 3.11.4.
4.1 Recycled, recovered materials.	Contractor certification that the recycled, recovered materials requirements are in accordance with 4.1.
4.2 Green Procurement Program (GPP).	Contractor certification that the GPP requirements are in accordance with 4.2.

5.5.2 Sound level test. A sound level test shall be performed on the defuel pumping system (if required) operations in accordance with the requirements as specified in 3.2.4.4. The maximum A-weighted sound levels produced by the truck during pumping operations shall not exceed 84 dBA at a distance of 15 feet from the geometric center of the truck cab and at the operator's position in front of the operator control panel. An additional 2 dBA allowance over this sound level limit shall be permitted in order to provide for variation in test site, temperature gradients, test equipment, and inherent differences in nominally identical vehicles. The test shall be performed on the first production model only.

5.5.3 Electromagnetic interference test. Certification for EMI testing shall be acceptable in lieu of testing. (See Table II).

5.5.4 Environmental testing. Certifications for environmental testing shall be acceptable in lieu of testing. (see Table II).

5.5.5 Weight and dimension tests.

5.5.5.1 Weight and center of gravity test. The weight, center of gravity, and axle weights of the first production HSV shall be measured to demonstrate compliance with the weight requirement of 3.4 and the axle weight requirement of 3.5.2.2. The test shall be performed on the first production model only.

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5.5.5.2 Dimension measurement. The first production HSV shall be measured to demonstrate compliance with the dimensional requirements of 3.4. The test shall be performed on the first production model only.

5.5.6 Transportability verification.

5.5.6.1 Air transportability analysis. Certification for Air transportability analysis shall be acceptable in lieu of an engineering analysis.

5.5.6.2 Equipment removal and reconfiguration demonstration. A first production HSV shall be configured for transport on C-130, C-5, and C-17 aircraft and then reconfigured for operation to demonstrate compliance with 3.5.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. The demonstration shall be performed on the first production model only.

5.5.6.3 Tie down provision verification.

5.5.6.3.1 Tie down provision analysis. Certification for tie downs shall be acceptable in lieu of an engineering analysis.

5.5.7 Flow control system (FCS) test. A FCS test shall be performed to verify the FCS is capable of controlling flow rates from 0 to a minimum of 750 + 0/-25 GPM through the ground discharge hose(s) and flow rates from 0 to 1000 + 0/-25 GPM through the lift platform hose(s) as required in 3.2.7.1. The test shall also verify the FCS does not allow the flow rate to exceed the rated flow of the fuel filtration system.

5.5.8 Pressure control system (PCS) test. A PCS test shall be performed to verify the PCS is in compliance with the requirements of 3.2.7.2. The test shall verify: (1) the primary PCS is capable of independently controlling the single-point nozzle pressure at the aircraft to a maximum of 52 +3 PSIG, (2) the secondary PCS is capable of independently controlling the single-point nozzle pressure at the aircraft to a maximum of 57 +3 PSIG when the primary PCS fails, (3) with a two second shutdown at the aircraft, the PCS is capable of limiting pressure surges at the aircraft to 120 PSIG at the discharge single-point nozzle(s), (4) the PCS is able to sense fuel pressure through some form of pressure compensator in order to simulate single-point nozzle pressure at the discharge single-point nozzle(s), and (5) the in-line pressure control valve has an opening rate adjustable from 2 to 25 seconds and closing rate adjustable from 10 to 30 seconds. The test shall verify: (1) a pressure relief system is provided to prevent pressure buildup in the pumping system after all fueling operations are complete and single-point nozzles properly stored, and (2) the pressure relief system does not allow backflow of fuel into the recovery tank when the single-point nozzle(s) are connected to the aircraft single-point receptacles.

5.5.9 Pilot control valve demonstration. A demonstration shall be performed to show all pilot control valves are in compliance with the requirements of 3.2.7. The demonstration shall show: (1) all flow and pressure control system pilot control valves are located so that a mechanic can observe the operator control panel gauges, operate the controls, and adjust the pilot control valves while standing on the ground, and (2) the necessary safety features are in place for the pilot control valves to prevent unauthorized adjustments to the pumping system.

5.5.10 Defuel system test. If equipped with a defuel system, a defuel system test shall be performed to verify the defuel system is in compliance with the requirements of 3.2.7.3. The test shall verify the HSV is capable of: (1) defueling aircraft through the rear ground discharge hoses and through the lift platform hoses, and (2) defueling at a flow rate of 175 GPM without the

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assistance of aircraft boost pumps and 300 ± 10 gpm with the assistance of aircraft boost pumps with a hydrant pit pressure of 160 pounds per square inch gauge (psig). The test shall verify: (1) the defuel suction line incorporates an 8-mesh low-pressure drop strainer that is easily removed for cleaning, and (2) the defuel system activation occurs when engaging the defuel mode selector switch located on the operator control panel.

5.5.11 Lift platform demonstration. If equipped with a lift platform, a test shall be performed to verify the lift platform is in compliance with the requirements of 3.2.13. The test shall show: (1) the lift platform has variable height capability, (2) the lift platform is located adjacent to the rear of the cab and provided with two discharge hoses, (3) the lift platform is capable of being lowered to the lowest position with the lift platform hoses connected to the aircraft, (4) the lift platform is equipped with: (a) aluminum guard rails with an entry gate and a toe board, (b) an access ladder or steps, (c) single point single-point nozzle pressure gauge, (d) lift controls and manual override, and (e) an emergency engine shut-off switch; (5) the lift platform has the capability to raise from the stowed position to the height required to allow a small human, as defined by MIL-STD-1472, to service any wide bodied aircraft at the under wing receptacle, (6) provisions are provided for easy entry and exit of the platform, (7) the lift platform is capable of rising and remaining at any selected height with two 250-pound operators onboard, (8) there are no visible signs of excessive lean, sway, or instability that would constitute an unsafe operating condition of the fully loaded platform in any position, (9) the lift platform is capable of being fully raised or lowered in not less than 15 seconds or more than 35 seconds, (10) a feature is provided to prevent rapid accidental lowering of the platform in the event of any component failure, (11) a lowering device (manual override) is provided that will override the main control and provide a smooth controlled lowering of the platform and is located in a convenient location accessible by the operator from the ground, (12) the wing stop control will stop the platform from ascending if the highest portion of the lift comes in contact with the aircraft wing, (13) the safety interlock sets the parking brakes when the lift platform is not in the full down position, (14) the lift platform has a minimum 2000 pound lift capacity, (15) the lift platform deck does not extend higher than 54 inches from the ground when in the stowed position, and (16) the lift platform incorporates an appropriately sized, weather protected, operator control panel that contains the necessary controls for proper operation of the lift platform.

5.5.12 Water-slug shut-off device demonstration. A demonstration shall be performed to verify the water-slug shutoff device is in compliance with the requirements of 3.2.8.3. The demonstration shall show (1) the shutoff device stops the flow of fuel within 10 seconds when the water in the vessel reaches a predetermined level, (2) resumes fuel flow within 30 seconds when the water is lowered to a predetermined level, and (3) fuel flow is terminated when water is manually introduced into the system.

5.5.13 Fuel recovery tank demonstration. A demonstration shall be performed to verify the fuel recovery tank is in compliance with the requirements of 3.2.7.2.3. The demonstration shall show: (1) the tank is of sufficient size (24 hours of operational use) to allow for thermal expansion of fuel and fuel from the fuel filtration system air eliminator, (2) the tank is equipped with relief valves and vents as necessary for safe operation, and (3) the tank is equipped with a manual drain valve to allow for complete draining of the tank. The demonstration shall show that fuel recovery tank is equipped with an automatic emptying system that will empty the fuel recovery tank during refueling and defueling servicing operations and incorporates a full tank shutdown system. The demonstration shall show the automatic emptying system: (1) does not utilize an

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electrically driven pump, (2) does not allow the backflow of fuel into the recovery tank when not in use, (3) returns the fuel up-stream of the fuel filtration system, (4) does not allow the dumping of fuel on the ground, and (5) operates only when the deadman control is actuated. The following features shall be verified: (1) the system shall incorporate an override momentary-on push button located on the operator control panel to allow the manual operation of the automatic emptying system, (2) a gauge is provided on the tank to monitor the level of fuel and is readable by the operator standing on the ground, (3) a high-level warning light, illuminating Red, is provided for the recovery tank and is located on the operator control panel, (4) the high-level recovery tank warning light illuminates whenever the fuel level in the recovery tank exceeds the predetermined level where the automatic return system is scheduled to activate, and (5) the high-level recovery tank warning light remains illuminated until product level drops below the return system activation point.

5.5.14 Hose reel demonstration. A demonstration shall be performed to verify the hose reels are in compliance with the requirements of 3.2.9.2 and 3.2.9.4. The demonstration shall show: (1) the hose reels are designed for servicing with any portion of the hose unwound, (2) the hose reels are able to rewind the entire flooded hose at a rate of 1.5 feet per second or greater, (3) the hose reels are of the top pull type, (4) the hose reel has features to prevent over-travel during deployment and rewind, (5) the hose reels are furnished with a manual crank for emergency rewinding, and (6) the hose reel is designed to prevent the last 1/2 turn of hose from unwinding from the reel. The demonstration shall verify the following: (1) stainless steel guide rollers with permanently lubricated metal bearings are provided to prevent damage to the hose while winding or unwinding, (2) all moving parts (drive chains, gears, etc.) are equipped with protective coverings to prevent injury to the operator and damage to hoses, (3) provisions are made for storing the crank on the HSV, (4) the hose reel drum size is in accordance with EI 1529, and (5) the rewind assembly incorporates a clutch and brake assembly.

5.5.15 Deadman control system (DCS) test. A demonstration shall be performed to verify the DCS is in compliance with 3.2.10. The test shall verify: (1) the system terminates fuel flow to and from the aircraft in accordance with the parameters as specified in NFPA 407, (2) deactivation of the deadman device terminates fuel flow by: (a) closing the hydrant pit valve, and (b) closing a valve downstream of the hydrant coupler. If an inline control valve is provided, it shall be utilized as the valve downstream of the hydrant coupler. The test shall verify the following: (1) the DCS is equipped with 70 feet of supply hose and is positioned so the control handle can be operated from the operator control panel, ground discharge hose reels, and raised lift platform, (2) the deadman control lever is operable by an operator wearing artic type mittens, (3) the electrical DCS (if provided), has been certified as intrinsically safe and incorporates a timer system for continuous operator input, (4) the electrical DCS (if provided), has provisions for storage of the cable, (5) the deadman hose is stored on a reel, (6) the reel holds at any deployed hose length and incorporates a non-manual rewind system, (7) a ball stop is provided on the deadman hose near the deadman control lever to prevent over wind of the deadman hose, and (8) deactivation of the deadman device when in the defuel mode of operation, returns engine speed to idle regardless of the auxiliary throttle position.

5.5.16 Auxiliary throttle demonstration. A demonstration shall be performed to verify the auxiliary throttle is in compliance with 3.11.1. The demonstration shall be performed in the defuel mode of operation and shall show: (1) the auxiliary throttle is adjustable to within 50 revolutions per minute (rpm) of the maximum rated engine speed or maximum pump speed, (2)

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the auxiliary throttle is capable of controlling flow, (3) the auxiliary throttle is capable of holding any speed setting, and (4) the auxiliary throttle incorporates an emergency return to idle feature.

5.5.17 Emergency engine shutoff demonstration. A demonstration shall be performed to verify the emergency engine shutoff is in compliance with 3.2.13. The demonstration shall show: (1) the emergency engine shut-off switch shuts down the HSV engine without utilizing any other control(s), and (2) the emergency engine shut-off switch is an on-off toggle switch marked by a red circle, color number 31136 of SAE-AMS-STD-595, at least one inch in diameter.

5.5.18 Differential pressure gauge demonstration. A demonstration shall be performed to verify the differential pressure gauge is in compliance with 3.2.11.6. The demonstration shall show: (1) the differential pressure gauge is a non-calibrating gauge that is capable of continuously monitoring the pressure drop across the filter coalescer stage during refuel servicing operations, and (2) the differential pressure gauge is capable of registering and holding the highest-pressure drop achieved during refuel servicing operations.

5.5.19 Emergency fuel shutoff demonstration. A demonstration shall be performed to verify the emergency fuel shut off is in compliance with the requirements of 3.2.11.4. The demonstration shall show upon activation of each control switch, the flow of fuel flow is terminated in any mode, regardless of whether the deadman device is activated or deactivated. The demonstration shall verify the HSV incorporates three independent switches for emergency fuel shutoff. The location of each switch shall be as follows: (a) one on the operator control panel, (b) one on the passenger's side of the HSV opposite the operator control panel, and (c) one on the operator control panel of the lift platform. The demonstration shall show: (1) the emergency fuel shut-off switch is a toggle switch that is either activated or push pull, (2) the emergency fuel shut-off switch is not recessed, and (3) the emergency fuel shut-off switch is manually operated.

5.5.20 Power take-off (PTO) demonstration. A demonstration shall be performed to verify the PTO is in compliance with the requirements of 3.7.1.1. The demonstration shall show: (1) the defuel system pump is driven through a transmission driven PTO or split shaft gearbox design, (2) the system is designed to either move the HSV or drive the pump; never both concurrently, (3) PTO engagement, from road to pump mode or from pump to road mode, occurs without gear clash or system shock loading, (4) the auxiliary throttle does not operate while the PTO is in road mode and the HSV accelerator pedal does not operate while the PTO is in pump mode, and (5) PTO engagement is controlled from the driver's seated position and shall not occur unless the selector is in neutral, the engine is at idle, and the HSV parking brake is engaged.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6.1 Ordering data. Acquisition documents must specify the following:

a. Type of classification (see 2).

b. Type of finish, as required (see 3.2.1.4).

c. Rustproofing, if required (see 3.2.1.4.1).

d. Jet fuel product markings, as required 3.2.2.1.4.

e. CT ground servicing hose for extreme climatic conditions, if required (see 3.2.9.1).

f. CT hydrant hose for extreme climatic conditions, if required (see 3.2.9.4).

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g. Type of hydrant coupler, as required (see 3.2.9.9).

h. Spare tire and wheel, if required (see 3.7.6.1).

i. Spare wheel, if required (see 3.7.6.2).

j. Additional moisture removing device, if required (see 3.9.2.1).

7. NOTES.

7.1 Source of documents.

7.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents, except for GOST 10227-86, are available online at https://quicksearch.dla.mil/. A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).

7.1.2 Federal Acquisition Regulation FAR. Copies of FAR documents may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/index.html.

7.1.3 American Society for Testing and Materials ASTM International. Copies of ASTM documents. Should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959. Electronic copies of ASTM standards may be obtained from https://www.astm.org.

7.1.4 North Atlantic Treaty Organization NATO. NATO documents may be obtained at http://nso.nato.int/nso or from NATO Standardization Office, NATO HQ, Blvd Leopold III, B-1110, Brussels Belgium.

7.1.5 Society of Automotive Engineers SAE. Application for copies of SAE documents should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096. Electronic copies of SAE standards may be obtained from https://www.sae.org/.

7.1.6 National Fire Protection Association NFPA. NFPA documents may be obtained at www.nfpa.org/ or from National Fire Protection Association 11 Tracy Drive Avon, MA 02322.

7.1.7 The Code of Federal Regulations (CFR). CFR documents. may be obtained at https://www.govinfo.gov/app/collection/cfr or obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.1.8 American Society of Mechanical Engineers ASME. ASME documents may be obtained at www.asme.org or from American Society of Mechanical Engineers, P.O. Box 2900, Fairfield NJ 07007.

7.1.9 Gammon Technical Products. Gammon Technical Products information may be obtained at https://www.gammontech.com/ or from Gammon Technical Products 2300 Highway 34, Manasquan, NJ 08736.

7.1.10 American Petroleum Institute API. API documents may be obtained at www.api.org or from American Petroleum Institute Monogram Program 1220 L Street, NW Washington, DC 20005-4070.

7.1.11 Energy Institute EI. EI documents may be obtained at http://www.energyinst.org or from Energy Institute, 61 New Cavendish Street, London, W1G 7AR, United Kingdom.

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7.1.12 National Institute of Standards and Technology NIST. NIST documents may be obtained at https://www.nist.gov or from NIST, 100 Bureau Drive, Stop 1070, Gaithersburg, MD 20899-1070.

7.1.13 Department of transportation DOT. Documents may be obtained from, https://www.transportation.gov/.

7.1.14 Environmental Safety and Occupational Health (ESOH) documents. Documents may be obtained from, https://www.dau.edu/acquipedia-article/environment-safety-and-occupational-health-esoh.

7.1.15 National Electrical Manufacturers Association. (NEMA). Documents can be obtained at https://www.nema.org/.

7.1.16 Environmental Protection Agency (EPA). Documents can be obtained at https://www.epa.gov/.

7.1.17 Tire and Rim Association. Documents can be obtained at https://www.us-tra.org/.

7.2 Definitions.

7.2.1 Common hand tools. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

7.2.2 Special tool. A tool that is not commercially and readily available from a source other than the HSV contractor.

7.2.3 Recycled, recovered materials. Materials collected and recovered from solid waste and reprocessed to become a source of raw materials, as compared to virgin raw materials. The components, pieces and parts incorporated in the HSV may be newly fabricated from recovered materials to the maximum extent practicable, provided the HSV produced meets all other requirements of this CID. Used, rebuilt or re-manufactured components, pieces and parts is not to be incorporated.

7.2.4 Type III Hydrant System. The type III system is an "on demand" type system which maintains a constant pressure of approximately 65 PSIG. When pressure starts to drop, additional pumps activate. Flow rates of up to 1200 GPM are possible with pressures ranging from 60 to 160 PSIG at the farthest hydrant outlet from the operating pump house. Aircraft servicing pressure is controlled by a control valve mounted in the hydrant pit. The control valve receives sensing pressure (fuel) and air pressure from the hydrant hose truck. Defueling is accomplished by using the pump on the hose truck to overcome the hydrant pressure.

7.2.5 Alternate fuel. A fuel that can be used in place of the primary fuel with a possible loss of efficiency, engine adjustments may be necessary. The use of an alternate fuel may result in a change of maintenance or overhaul cost.

7.2.6 Primary fuel. The fuel or fuels used during tests to demonstrate system performance (contract compliance). Substitutions are presented as Alternate and Emergency fuels.

7.4 Key words.

Air transportable

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Commercial aircraft

Hydrant system

Military aircraft

MILITARY INTERESTS

Custodian:
Air Force – 184

Preparing activity:
Air Force – 184

Agent:
Air Force – 110

(Project 2320-2024-001)

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